Assignment 2

1(a.) $\neg q \to \neg p$ ("if bella does not have teeth, she is not a cat")

B.) $\neg p \to \neg q$ ("if Bella is not a cat, then she does not have teeth.")

C.) "if Bella has teeth, she is a cat."
Proof: if Bella has teeth, she could also be a human. ∴ this is not valid.

2.) $((p \vee q) \wedge (r \to q) \wedge (q \to \neg r)) \to p$

p	q	r	$p \vee q$	$r \to q$	$q \to \neg r$	LHS	$\to p$	LHS $\to p$
1	1	1	1	1	0	0		1
1	1	0	1	1	1	1		1
1	0	1	1	0	1	0		1
1	0	0	1	1	1	1		1
0	1	1	1	1	0	0		1
0	1	0	1	1	1	1		$\boxed{0}$ 6.
0	0	1	0	0	1	0		1
0	0	0	0	1	1	0		1

As its LHS $\to p$, for all p it will be true. ∴ the only rows that matter are rows where p is not true (5, 8, 6, 7) or for when LHS is true (2, 3, 5)

b.) No, its not, row 6 $\to 0$

3a.) false, $5^2(5-6) = -25$, $6 - 11 \times 5 = -49$ ∴ $-25 \neq -49$

b.) true, if n is an even integer then $n = 2k$ is $11(2k) = 22k = 2(11k)$, where if k is any integer, then $2(k)$ is always even.

C.) for odd numbers it is correct. The statement specify's all integers (including even) and is an "or" statement. ∴ if we can prove n^2 is even for even no. then the statement is true. if $n = 2k$, then $n^2 = (2k)^2 = 2(2k^2)$. ∴ the statement is true for even and odd